Trigonometric Approach to IK

Summary:

This lesson introduces inverse kinematics (IK), the reverse of the forward kinematics from Lessons 3–5, and solves it with trigonometry. On Day 1, students use the law of cosines to find the elbow angle E and the shoulder angle S of the simplified 2-link planar arm for a given target (X, Y). On Day 2, students extend the method to the 3D arm: the base angle comes from atan2, the target is flattened into the arm's vertical plane, and Day 1's steps finish the job. By the end of the lesson, students can calculate the joint angles (B, S, E) for a reachable target (x, y, z), decide when a target is out of reach, and check their answer with the forward kinematics of Lesson 4.

Objective:

By the end of this lesson, students will be able to:

- Use the law of cosines to find an angle of a triangle when all three sides are known
- Explain why the two links and the line to the target form a triangle, and use it to find the elbow angle E and the shoulder angle S of the 2-link planar arm
- Find the base angle B with atan2 and flatten a 3D target into the arm's plane
- Decide whether a target is reachable, and explain why a target can have two solutions (elbow up and elbow down)
- Verify an IK answer with the forward kinematics of Lesson 4, and discuss the strengths and limits of the trigonometric method

Standards:

NGSS

- Engineering Design: HS-ETS1-2 (break a complex problem into smaller problems), HS-ETS1-3 (evaluate solutions against trade-offs), HS-ETS1-4 (use a computer model to test a solution)
- Science & Engineering Practices: Developing and Using Models; Using Mathematics and Computational Thinking; Engaging in Argument from Evidence
- Crosscutting Concepts: Systems and System Models; Patterns

Vocabulary:

Inverse kinematics (IK): Finding the joint angles that put the end effector at a given target position. It is the reverse of forward kinematics.

Target: The (x, y, z) position we want the end effector to reach.

Law of cosines: A rule for any triangle: $c^2 = a^2 + b^2 - 2ab \cos C$. It lets us find an angle when we know all three sides.

Inverse cosine ($\cos^{-1}$, arccos): The calculator key that turns a cosine value back into its angle. Used in degrees mode.

D: The straight-line distance from the shoulder to the target. With the two links it forms a triangle: $D = \sqrt{(X^2 + Y^2)}$ for the flat arm.

ϕ and δ: Two angles used for the shoulder: $\phi = \tan^{-1}(Y \div X)$ is the tilt of the line D, and δ is the angle between D and the upper arm, found with the law of cosines.

atan2(y, x): A version of $\tan^{-1}$ that looks at the signs of both y and x, so it gives the correct quadrant. It is used for the base angle B.

Horizontal reach r and height h: The target's distance from the base axis, $r = \sqrt{(x^2 + y^2)}$, and its height, $h = z$. They flatten the 3D problem into the 2D problem of Day 1.

Elbow up / elbow down: The two arm shapes that usually reach the same target. They differ in the sign of E.

Reachable: A target the arm can reach: $D \le L_1 + L_2 = 500$ mm. If $|\cos E| > 1$, the target is out of reach.

Resources & Materials:

- Lesson 6 slideshow, Days 1 and 2 (34 slides; the slide numbers in the Procedures refer to it)
- Calculators with $\cos^{-1}$ and $\tan^{-1}$ keys, set to degrees mode, for the Try It Yourself problems
- The assembled 4-DOF arm from Lesson 2, to show reach and the elbow-up and elbow-down shapes (optional)
- Computers with Python and NumPy, plus the helper files ik_helper.py (code in the Appendix) and dh_helper.py (from Lesson 4), for the Check It with Code slide

Procedures:

Day 1: IK of the Flat Arm with Triangles

- **Opening (4 min)** -- Display the title slide, then "Where Lesson 5 Left Off" and "Forward vs. Inverse Kinematics" (slides 1–3). Recall that FK turned (B, S, E) = (45°, 45°, 45°) into the tip position (301.78, 301.78, 176.78). Today we go backwards: the robot is given a target and needs the angles. Lesson 3's triangles are used again, run in reverse. Remind students of the notation: z is up, S is measured clockwise from vertical, and E = 0° is a straight arm.
- **The IK problem (4 min)** -- Show "The IK Problem (Flat Arm)" (slide 4). Given $L_1 = L_2 = 250$ mm and a target (X, Y), find S and E. Solve the flat arm first; the base turn is added on Day 2.
- **Law of cosines (7 min)** -- Show "The Law of Cosines" (slide 5): $c^2 = a^2 + b^2 - 2ab \cos C$, and the rearranged form $\cos C = (a^2 + b^2 - c^2) \div (2ab)$. Point out that C = 90° gives Pythagoras. Work through the 250, 250, 300 triangle on "Worked Example: Law of Cosines" (slide 6): cos C = 0.28, so C = 73.74°. Check that calculators are in degrees mode.
- **Try It Yourself #1 (5 min)** -- Students find the angle opposite c for sides 250, 250, 250 and 250, 250, 400 (slide 7). Reveal solutions 60° and 106.26°. Use (b) to show that a negative cosine gives an angle larger than 90°.

- **The arm makes a triangle (6 min)** -- Show "The Arm Makes a Triangle" (slide 8). The two links and the line D from the shoulder to the target form a triangle in which all three sides are known: L_1, L_2 and $D = \sqrt{(X^2 + Y^2)}$. The angles of this triangle are the joint angles.
- **Step 1: the elbow (8 min)** -- Show "Find the Elbow Angle E" (slide 9). The law of cosines gives the inside angle γ, but E is the bend on the outside, so $E = 180° - \gamma$, which gives $\cos E = (D^2 - L_1^2 - L_2^2) \div (2 L_1 L_2)$. Check it with a straight arm (D = 500, E = 0°). Then work "Worked Example: The Elbow" (slide 10) for (426.78, 176.78): $D^2 = 213{,}389$, cos E = 0.7071, E = 45°. Common mistake: using γ as E.
- **Step 2: the shoulder (8 min)** -- Show "Find the Shoulder Angle S" (slide 11): the upper arm is $\phi + \delta$ above the horizontal, where $\phi = \tan^{-1}(Y \div X)$ and $\cos \delta = (L_1^2 + D^2 - L_2^2) \div (2 L_1 D)$. Then $S = 90° - (\phi + \delta)$, the same conversion as in Lesson 4. Work "Worked Example: The Shoulder" (slide 12): $\phi = 22.5°$, $\delta = 22.5°$, S = 45°.
- **Completed variables (3 min)** -- Show "Completed Variables" (slide 13). IK recovered the angles that Lesson 4 started with (S = 45°, E = 45°). Putting them back into Lesson 4's chain returns (426.78, 176.78).
- **Try It Yourself #2 and #3 (7 min)** -- Students solve (341.51, 341.51) and (250, 250) (slides 14–15). Reveal solutions S = 30°, E = 30°, and S = 0°, E = 90°. The first target is Lesson 4's Try It Yourself #2 run backwards, so students can compare.
- **Elbow up or elbow down (3 min)** -- Show "Elbow Up or Elbow Down" (slide 16). The arccos gives only a positive angle, so we choose the sign of E. For elbow up, $a = \phi + \delta$; for elbow down, E is negative and $a = \phi - \delta$. For (250, 250), elbow down gives E = −90°, S = 90°. Teacher note: check the arm's joint limits before using elbow down. This lesson uses elbow up.
- **Reachability (3 min)** -- Show "Can the Arm Reach It?" (slide 17). A triangle needs $D \le L_1 + L_2 = 500$ mm. If $|\cos E|$ is bigger than 1, $\cos^{-1}$ fails and the target is out of reach. Always check cos E first.
- **Recap & wrap-up (2 min)** -- Show the "Day 1 Recap" slide (slide 18): triangle, elbow, shoulder, and checks. Preview Day 2: the real arm's base turns too, so we need 3D.

Day 2: Extending to 3D

- **Recap (3 min)** -- Show "Extending to 3D" (slide 19) and review the Day 1 steps. Quick-fire questions: What is E for a straight arm? (0°.) What is the largest D the arm can reach? (500 mm.)
- **The base aims the arm (5 min)** -- Show "The Base Aims the Arm" (slide 20). Viewed from above, the base turns until the arm points at the target's (x, y) spot: $B = \tan^{-1}(y \div x)$. For (301.78, 301.78), B = 45°.
- **Flatten 3D into 2D (4 min)** -- Show "Flatten 3D Into 2D" (slide 21). The arm lives in one vertical plane. Measure the target inside that plane: $r = \sqrt{(x^2 + y^2)}$ and h = z. Then (r, h) plays the role of (X, Y) from Day 1, and the elbow and shoulder steps are unchanged.
- **Why atan2? (5 min)** -- Show "Why atan2?" (slide 22). For (−125, 216.51), $\tan^{-1}(y \div x) = -60°$, which points the wrong way, while atan2 gives 120°. Teacher note: atan2 returns angles from −180° to 180°. If the servo range is 0° to 360°, add 360° to a negative B. Also point out that x = y = 0 has no single B.
- **The full recipe (4 min)** -- Show "The Full Recipe" (slide 23): (1) B = atan2(y, x); (2) $r = \sqrt{(x^2 + y^2)}$, h = z, $D^2 = r^2 + h^2$; (3) $\cos E = (D^2 - L_1^2 - L_2^2) \div (2 L_1 L_2)$, after checking $|\cos E| \le 1$; (4) $S = 90° - (\phi + \delta)$.
- **Worked example (12 min)** -- Walk through the target (301.78, 301.78, 176.78) on "Worked Example: Setup", "Aim and Flatten" and "Elbow and Shoulder" (slides 24–26). B = 45°, r = 426.78, h = 176.78,

which is Day 1's worked example, so E = 45° and S = 45°. Students should predict B = S = E = 45° before the calculation, because the point came from Lesson 4.

- **Check with Lesson 4 (3 min)** -- Show "Check It with Lesson 4" (slide 27). Feed the IK angles into the DH table (θ = 45°, 90° − 45° = 45°, −45°) and see that FK lands back on the target. Make this a habit: always check IK with FK.
- **Try It Yourself #4 (5 min)** -- Students solve (187.50, 324.76, 216.51) (slide 28); reveal B = 60°, S = 30°, E = 60°. Here r = 375.00 and h = 216.51.
- **Try It Yourself #5 (4 min)** -- Students solve (−324.76, 187.50, 216.51) (slide 29); reveal B = 150°, S = 30°, E = 60°. It has the same r and h as #4, with the base turned 90° further. Plain $\tan^{-1}$ would say −30°. If time is short, assign this problem as homework.
- **Try It Yourself #6 (3 min)** -- Students decide whether (200, 200, 200) and (400, 300, 200) are reachable (slide 30). Reveal: D = 346.41 is reachable; D = 538.52 is out of reach (cos E = 1.32).
- **Check it with code (4 min)** -- Run the code on "Check It with Code" (slide 31), which prints 45.0 45.0 45.0 and then [301.78, 301.78, 176.78]. Students change the target to the values from Try It Yourself #4–#6 to check their answers. The helper files do the work; students write IK code themselves later in the course.
- **When IK gets tricky (3 min)** -- Show "When IK Gets Tricky" (slide 32). FK always gives one answer; IK can give none (D > 500), one (D = 500), two (elbow up or down), or infinitely many (x = y = 0). Near full reach, rounding in the target changes E a lot (for example, (433.01, 250.00) gives E = 0.35°, not 0°), so allow a small tolerance.
- **Trigonometry for IK (3 min)** -- Show "Trigonometry for IK: Pros and Cons" (slide 33). Ask: if the arm had six joints, which method would you use for IK, and why? Trigonometry gives an exact, fast answer for our arm, but each new joint needs new triangles. If the lesson runs long, shorten this to a one-minute summary.
- **Wrap-up (2 min)** -- Show "What We Learned" (slide 34): IK goes from target to angles; the arm makes a triangle, so the law of cosines gives E and δ; B = atan2(y, x), with r flattening 3D into 2D; S = 90° − (ϕ + δ); and check the result with FK. Preview Lesson 7: gradient descent IK.

Assessment:

Complete the Try It Yourself problems in the slideshow (#1–#6). Students check each answer against the solution on the slide and can verify #4–#6 with the code on the "Check It with Code" slide.

Use these questions as an exit ticket or short quiz. The answers are in red.

1. Find the angle opposite the side of length 350 in a triangle with sides 250, 250 and 350.
   Answer: cos C = (125,000 − 122,500) ÷ 125,000 = 0.02, so C = 88.85°.
2. The arm has $L_1 = L_2 = 250$ mm. Can it reach (400, 300, 200)? Show why or why not.
   Answer: No. r = 500 and h = 200, so D = 538.52 mm, which is more than 500 mm. cos E would be 1.32, which is impossible.
3. Find S and E for the flat-arm target (X, Y) = (0, 500).
   Answer: D = 500, so cos E = 1 and E = 0°. ϕ = 90°, δ = 0°, so S = 90° − 90° = 0°. The arm is straight up.

4. Why do we use atan2(y, x) instead of $\tan^{-1}(y \div x)$ for the base angle B?
   Answer: $\tan^{-1}$ cannot tell opposite quadrants apart (and fails when x = 0). atan2 uses the signs of both y and x, so it gives the correct angle, such as 120° for (−125, 216.51).
5. True or False: every target the arm can reach has exactly one set of joint angles.
   Answer: False. Most targets have two (elbow up and elbow down). Only a fully straight arm (D = 500 mm) has one, and a target on the base axis has infinitely many values of B.
6. List the four steps of the trigonometric IK recipe.
   Answer: (1) B = atan2(y, x). (2) $r = \sqrt{(x^2 + y^2)}$, h = z. (3) Elbow: $\cos E = (D^2 - L_1^2 - L_2^2) \div (2\, L_1\, L_2)$. (4) Shoulder: $S = 90° - (\phi + \delta)$.
7. Find B, S and E for the target (0, 250, 250).
   Answer: B = atan2(250, 0) = 90°. r = 250 and h = 250, so E = 90° and S = 0°.
8. How can you check an IK answer?
   Answer: Put B, S and E into Lesson 4's DH matrices and multiply. The last column of T should match the target.
9. In one or two sentences, why is trigonometry a good choice for our arm but not for a six-joint arm?
   Answer: For our arm the triangles give an exact, fast answer. A six-joint arm needs new triangles and many special cases for every extra joint, so a method that does not depend on drawing triangles works better.

Appendix: ik_helper.py

Save this code as ik_helper.py in the same folder as dh_helper.py. The "Check It with Code" slide imports ik_trig from it.

```
"""Helper functions for Lesson 6: the trigonometric approach to IK.

Angles are in degrees. Lengths are in mm.
Conventions (same as Lesson 4): z points up, B turns about z, S is measured
clockwise from vertical, E is the clockwise bend of the elbow (0 = straight).
"""
import numpy as np

L1 = 250  # upper arm
L2 = 250  # forearm


def ik_2d(X, Y, elbow_up=True):
    """Flat arm: target (X, Y) -> (S, E). Raises ValueError if out of reach."""
    D2 = X**2 + Y**2
    cosE = (D2 - L1**2 - L2**2) / (2 * L1 * L2)
    if abs(cosE) > 1.001:  # small tolerance: rounded targets can sit right at full
reach
        raise ValueError("target is out of reach")
    E = np.arccos(np.clip(cosE, -1, 1))
    if not elbow_up:
        E = -E
    phi = np.arctan2(Y, X)                                  # angle of the target line
    delta = np.arctan2(L2 * np.sin(E), L1 + L2 * np.cos(E))  # upper arm vs. that line
    a = phi + delta
    S = 90 - np.degrees(a)
    return float(S), float(np.degrees(E))


def ik_trig(x, y, z, elbow_up=True):
    """3D arm: target (x, y, z) -> (B, S, E)."""
    B = np.degrees(np.arctan2(y, x))
    r = np.hypot(x, y)
    S, E = ik_2d(r, z, elbow_up)
    return float(B), S, E


if __name__ == "__main__":
    # The Lesson 6 worked example: the tip from Lesson 4.
    B, S, E = ik_trig(301.78, 301.78, 176.78)
    print([round(v, 2) + 0.0 for v in (B, S, E)])  # expected: [45.0, 45.0, 45.0]
```